RKG Wallah Academy

Class 12 Chemistry — Complete Notes

Chapter-wise notes: Solutions, Electrochemistry, Kinetics, d/f-block, Coordination, Organic and Biomolecules.

Note: Ye NCERT-based summary notes hain. Diagrams NCERT se apni copy mein banao aur exam se pehle apni book se ek baar mila lo.

1. Solutions

• Solution: homogeneous mixture of two or more components. Solvent is in larger amount.

• **Concentration terms:** mass % = (mass of solute/mass of solution)×100; ppm; mole fraction x; Molarity M = moles of solute/L of solution (changes with temperature); Molality m = moles of solute/kg of solvent (independent of temperature).

Mole fraction $x_1 = n_1/(n_1+n_2)$ $x_1 + x_2 = 1$

• **Henry's law:** partial pressure of a gas in vapour phase is proportional to its mole fraction in solution: $p = K_H x$. Higher K_H means lower solubility. Solubility of gases decreases with temperature.

• **Raoult's law:** for a solution of volatile liquids, $p_1 = x_1 p_1^\circ$ and $p_2 = x_2 p_2^\circ$; total $p = p_1 + p_2$. For a non-volatile solute: relative lowering of vapour pressure $(p^\circ - p)/p^\circ = x_2$.

• Ideal solutions obey Raoult's law ($\Delta H_{mix} = 0$, $\Delta V_{mix} = 0$). Positive deviation: weaker A–B interactions (ethanol + acetone), minimum-boiling azeotrope. Negative deviation: stronger A–B interactions (HNO_3 + water), maximum-boiling azeotrope.

Colligative properties

Depend only on the number of solute particles, not on their nature: relative lowering of vapour pressure, elevation of boiling point, depression of freezing point and osmotic pressure.

$\Delta T_b = K_b m$ $\Delta T_f = K_f m$ $\pi = CRT$ $M_2 = (K \cdot w_2 \times 1000)/(\Delta T \times w_1)$ (w_1 in g solvent)

• Osmosis: flow of solvent through a semipermeable membrane towards the higher solute concentration. Reverse osmosis (RO) at pressure above π is used for desalination.

• **van't Hoff factor i** = normal molar mass / observed molar mass = observed colligative property / calculated. For dissociation (NaCl, i ≈ 2), association (ethanoic acid in benzene, i ≈ 0.5). Modified: $\Delta T_b = iK_b m$, $\pi = iCRT$.

• K_b (water) = 0.52 K kg/mol; K_f (water) = 1.86 K kg/mol.

Q. Find the freezing point of a solution of 18 g glucose (M = 180) in 500 g water. (K_f = 1.86)

Sol. m = (18/180)/0.5 = 0.2 mol/kg. $\Delta T_f = 1.86 \times 0.2 = 0.372$ K. Freezing point = **−0.372 °C**.

2. Electrochemistry

• **Galvanic (voltaic) cell:** converts chemical energy to electrical energy via spontaneous redox. Anode: oxidation (−); cathode: reduction (+). Daniell cell: Zn | Zn^{2+} || Cu^{2+} | Cu, $E°_{cell}$ = 1.10 V. Salt bridge maintains neutrality.

• **Electrode potential** (standard: 1 M, 298 K, 1 bar). Standard hydrogen electrode (SHE) = 0.00 V.

$E°_{cell} = E°_{cathode} - E°_{anode}$ $\Delta G° = -nFE°$ F = 96500 C/mol

Nernst equation (298 K): $E_{cell} = E°_{cell} - (0.059/n) \log Q$ At equilibrium: $\log K_c = nE°/0.059$

• **Conductance:** G = 1/R (S). Conductivity κ = G × (l/A) (S cm^{-1}). Molar conductivity Λ_m = κ × 1000/M (S cm^2 mol^{-1}); increases on dilution.

• **Kohlrausch law:** limiting molar conductivity of an electrolyte = sum of the limiting ionic conductivities: $\Lambda°_m = \nu_+\lambda°_+ + \nu_-\lambda°_-$. For weak electrolytes: degree of dissociation α = $\Lambda_m/\Lambda°_m$.

• **Faraday's laws of electrolysis:** (1) mass deposited w = Z I t (Z = electrochemical equivalent); (2) masses deposited by same charge are proportional to equivalent masses.

• **Batteries:** primary (dry cell, mercury cell – not rechargeable); secondary (lead storage battery, Ni–Cd – rechargeable). Fuel cell: H_2–O_2, efficiency ~70%, pollution-free.

• **Corrosion** (rusting of iron) is an electrochemical process; prevent by painting, galvanising (zinc coat), cathodic protection (sacrificial anode).

3. Chemical Kinetics

• **Rate of reaction:** change in concentration per unit time. Average and instantaneous rate. Rate law: Rate = $k[A]^x[B]^y$; order = x + y (experimental); molecularity (elementary steps only, whole numbers 1, 2, 3).

	Zero order	First order
Integrated rate law	$[R] = [R]_0 - kt$	$k = (2.303/t) \log([R]_0/[R])$
Half-life	$t_{½} = [R]_0/2k$	$t_{½} = 0.693/k$ (independent of concentration)
Unit of k	mol L^{-1} s^{-1}	s^{-1}
Graph	[R] vs t straight line	ln[R] vs t straight line

• **Arrhenius equation:** $k = A\, e^{-E_a/RT}$. Activation energy E_a is the minimum extra energy for a reaction. Rate roughly doubles for every 10 K rise (temperature coefficient 2–3).

$\log(k_2/k_1) = (E_a/2.303R)(1/T_1 - 1/T_2)$ R = 8.314 J/mol K

• **Collision theory:** reaction occurs only by effective collisions having sufficient energy (threshold) and proper orientation. Catalyst lowers E_a and speeds up the reaction without being consumed.

4. The d- and f-Block Elements

• Transition elements: general configuration $(n-1)d^{1-10}\ ns^{1-2}$. Zn, Cd, Hg are not considered transition elements (full d^{10}).

• Properties: high melting point (strong metallic bonding), variable oxidation states, coloured ions (d–d transition), paramagnetism (unpaired electrons), catalytic activity, complex formation, interstitial compounds, alloys.

• Magnetic moment $\mu = \sqrt{[n(n+2)]}$ BM (n = number of unpaired electrons).

• **$KMnO_4$:** prepared from pyrolusite (MnO_2) → K_2MnO_4 (green) → $KMnO_4$ (purple). Strong oxidising agent: in acidic medium $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$.

• **$K_2Cr_2O_7$:** orange; in acidic medium $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$. Chromate (yellow, CrO_4^{2-}) ⇌ dichromate (orange) depends on pH.

• **Lanthanoids** (4f series, Ce–Lu): common oxidation state +3; **lanthanoid contraction** = steady decrease in ionic size with increasing atomic number (poor shielding by 4f electrons), making 4d and 5d elements similar in size.

• **Actinoids** (5f series, Th–Lr): all radioactive; show variable oxidation states (+3, +4, +5, +6).

5. Coordination Compounds

• Complex: central metal atom/ion bound to ligands. **Coordination number** = number of ligand donor atoms attached to the metal. **Coordination sphere** is written in square brackets.

• Ligands: unidentate (Cl^-, NH_3, H_2O, CN^-), bidentate (ethylenediamine en, oxalate), polydentate (EDTA, hexadentate), ambidentate (NO_2^-, SCN^-).

• **Werner's theory:** primary valency (oxidation state, ionisable) and secondary valency (coordination number, non-ionisable).

• **IUPAC naming:** ligands first (alphabetical), then metal; oxidation state in Roman numeral; anionic complex ends with "-ate". Example: $[Co(NH_3)_6]Cl_3$ = hexaamminecobalt(III) chloride; $K_4[Fe(CN)_6]$ = potassium hexacyanoferrate(II).

• **Isomerism:** structural (ionisation, linkage, coordination, hydrate) and stereoisomerism (geometrical cis–trans, optical).

• **Valence bond theory:** inner orbital (d^2sp^3, low spin, strong ligand, e.g. $[Fe(CN)_6]^{3-}$) vs outer orbital (sp^3d^2, high spin, weak ligand, e.g. $[FeF_6]^{3-}$). Tetrahedral: sp^3; square planar: dsp^2.

• **Crystal field theory:** octahedral splitting into t_2g (lower) and e_g (higher); Δ_o energy gap. Strong field ligands give low spin (Δ_o > pairing energy). Colour arises from d–d transitions. Spectrochemical series: $I^- < Br^- < Cl^- < F^- < OH^- < H_2O < NH_3 <$ en $< CN^- <$ CO.

• Uses: extraction of metals, EDTA in water hardness, cisplatin in cancer treatment, chlorophyll (Mg), haemoglobin (Fe), vitamin B_{12} (Co).

6. Haloalkanes and Haloarenes

• Classification: alkyl (primary, secondary, tertiary), allylic, benzylic, vinylic, aryl halides.

• **Preparation:** from alcohols ($SOCl_2$ best, $HCl/ZnCl_2$, PCl_5); free-radical halogenation; Finkelstein (R–Cl + NaI → R–I); Swarts (R–Cl + AgF → R–F); Sandmeyer (diazonium salt + CuCl/CuBr).

• Boiling point increases with mass and decreases with branching; haloalkanes are insoluble in water.

	SN1	SN2
Mechanism	Two steps, carbocation intermediate	One step, concerted, backside attack
Rate	Rate = k[RX] (first order)	Rate = k[RX][Nu] (second order)
Order of reactivity	3° > 2° > 1°	1° > 2° > 3°
Stereochemistry	Racemisation	Inversion (Walden)
Solvent	Polar protic	Polar aprotic

• Elimination: with alcoholic KOH gives alkene (Saytzeff rule: more substituted alkene is major).

• Wurtz: 2R–X + 2Na → R–R. Fittig and Wurtz–Fittig for aromatic.

• Grignard reagent: R–X + Mg (dry ether) → RMgX.

• Haloarenes are less reactive in nucleophilic substitution because of resonance and sp^2 C–X partial double bond character; electron-withdrawing groups (NO_2) at ortho/para increase reactivity.

• Uses/Environment: chloroform (anaesthetic, forms phosgene in air), CCl_4, freons (ozone depletion), DDT (non-biodegradable).

7. Alcohols, Phenols and Ethers

• Alcohols: R–OH; phenols: Ar–OH; ethers: R–O–R′. Alcohols classified as 1°, 2°, 3°.

• **Preparation of alcohols:** hydration of alkenes; hydroboration–oxidation (anti-Markovnikov); reduction of aldehydes/ketones ($NaBH_4$, $LiAlH_4$); Grignard reagent + carbonyl; fermentation of sugars to ethanol.

• **Phenol:** from cumene (cumene hydroperoxide → phenol + acetone), from diazonium salt, from chlorobenzene (Dow).

• **Properties:** alcohols form H-bonds, high boiling points. Acidity: phenol > water > alcohols; phenol is acidic due to stabilised phenoxide ion. Electron-withdrawing groups raise acidity.

• **Distinguishing tests:** Lucas test ($ZnCl_2$ + conc. HCl): 3° immediately turbid, 2° in 5 min, 1° no turbidity at room temperature. Phenol + $FeCl_3$ gives violet colour.

• Reactions: with Na give H_2; dehydration (H_2SO_4 at 443 K → alkene; at 413 K → ether); oxidation (1° → aldehyde (PCC) → acid; 2° → ketone; 3° resists); esterification.

• **Phenol reactions:** electrophilic substitution (ortho/para directing, activated ring): bromine water gives 2,4,6-tribromophenol (white ppt); nitration; **Reimer–Tiemann** ($CHCl_3$/NaOH → salicylaldehyde); **Kolbe** (NaOH, CO_2 → salicylic acid).

• **Ethers:** Williamson synthesis R–X + R′ONa → R–O–R′ (best with 1° halide). Cleavage with HI gives alcohol + alkyl iodide. Ethers are inert, used as solvents.

8. Aldehydes, Ketones and Carboxylic Acids

• Carbonyl group C=O. Aldehydes: R–CHO; ketones: R–CO–R′; acids: R–COOH.

• **Aldehyde preparation:** oxidation of 1° alcohols; ozonolysis; **Rosenmund** (acyl chloride + H_2/Pd–$BaSO_4$); **Stephen** (nitrile + $SnCl_2$/HCl, then H_3O^+); **Etard** (toluene + CrO_2Cl_2); **Gattermann–Koch** (benzene + CO/HCl, $AlCl_3$/CuCl → benzaldehyde).

• **Nucleophilic addition** is the characteristic reaction (aldehydes more reactive than ketones: less steric hindrance, more positive carbon). Addition of HCN gives cyanohydrin; $NaHSO_3$ gives bisulphite compound; NH_2-Z derivatives give oximes, hydrazones, 2,4-DNP derivative (orange ppt).

• **Aldol condensation:** aldehydes/ketones with α-H, dilute alkali → β-hydroxy carbonyl → α,β-unsaturated carbonyl on heating. **Cannizzaro:** aldehydes without α-H (HCHO, benzaldehyde) with conc. alkali → alcohol + acid salt.

• Reductions: $LiAlH_4$/$NaBH_4$ → alcohol; **Clemmensen** (Zn–Hg/HCl) and **Wolff–Kishner** (NH_2NH_2, KOH/glycol) → alkane.

• **Tests: Tollens'** (ammoniacal $AgNO_3$): aldehydes give silver mirror; **Fehling's**: aliphatic aldehydes give red-brown Cu_2O ppt (aromatic aldehydes do not); **Iodoform test**: methyl ketones (CH_3CO–) and ethanol give yellow ppt of CHI_3.

• **Carboxylic acids:** acidic due to resonance-stabilised carboxylate; acidity increases with electron-withdrawing groups (e.g., $ClCH_2COOH$ > CH_3COOH) and decreases with EDG. Form dimers (high b.p.).

• Preparation: oxidation of 1° alcohols/aldehydes, hydrolysis of nitriles, Grignard + CO_2. Reactions: esterification, reaction with PCl_5/$SOCl_2$ giving acyl chlorides, decarboxylation (soda lime), **Hell–Volhard–Zelinsky (HVZ)**: α-halogenation using Cl_2/Br_2, red P.

9. Amines

• Amines are derivatives of NH_3: 1°, 2°, 3°. Nomenclature: ethanamine, N-methylethanamine, aniline (benzenamine).

• **Preparation:** reduction of nitro compounds (Sn/HCl, H_2/Pd); ammonolysis of alkyl halides; reduction of nitriles/amides; **Gabriel phthalimide** synthesis (only 1° amines); **Hofmann bromamide** (amide + Br_2 + NaOH → 1° amine with one carbon less).

• **Basicity:** in aqueous solution for ethyl amines 2° > 3° > 1° > NH_3; for methyl amines 2° > 1° > 3° > NH_3. Aromatic amines are weaker bases than aliphatic (lone pair delocalised into ring).

• **Reactions:** acylation, alkylation, **carbylamine test** (1° amine + $CHCl_3$ + KOH → foul-smelling isocyanide), **Hinsberg's test** (benzenesulphonyl chloride: 1° gives product soluble in alkali; 2° insoluble; 3° no reaction), nitrous acid: 1° aliphatic gives alcohol + N_2; 1° aromatic gives diazonium salt at 273–278 K.

• Aniline: bromine water gives 2,4,6-tribromoaniline; to get mono substitution protect $-NH_2$ by acetylation. Nitration, sulphonation (sulphanilic acid).

• **Diazonium salts** ($ArN_2^+X^-$): Sandmeyer (CuCl, CuBr, CuCN), Gattermann (Cu powder), replacement by I (KI), F (HBF_4 — Balz–Schiemann), OH (H_2O, warm), H (H_3PO_2). **Coupling reaction** with phenol/aniline gives azo dyes (coloured, −N=N−).

10. Biomolecules

• **Carbohydrates:** polyhydroxy aldehydes/ketones. Monosaccharides (glucose, fructose, galactose), disaccharides (sucrose = glucose + fructose; maltose = glucose + glucose; lactose = glucose + galactose), polysaccharides (starch, cellulose, glycogen).

• **Glucose:** $C_6H_{12}O_6$, an aldohexose; exists as cyclic pyranose with α and β anomers. Sucrose is non-reducing. Starch = amylose (linear, α-1,4) + amylopectin (branched); cellulose is β-glucose polymer; glycogen is animal starch.

• **Proteins:** polymers of α-amino acids joined by **peptide bonds** (–CO–NH–). 20 amino acids; essential amino acids cannot be made by the body. Zwitterion form; isoelectric point.

• Structure levels: primary (sequence), secondary (α-helix, β-pleated sheet), tertiary (3-D folding), quaternary (subunits). Denaturation: loss of structure and biological activity (boiling egg).

• **Enzymes:** biocatalysts, mostly proteins, specific (lock-and-key). **Vitamins:** fat-soluble (A, D, E, K) and water-soluble (B, C). Deficiency: A – night blindness; B_1 – beriberi; B_{12} – pernicious anaemia; C – scurvy; D – rickets; K – blood clotting problem.

• **Nucleic acids:** DNA (deoxyribose; A, G, C, T) and RNA (ribose; A, G, C, U). Nucleotide = base + sugar + phosphate. DNA is double helix with A=T (2 H-bonds) and G≡C (3 H-bonds); stores genetic information. RNA types: mRNA, rRNA, tRNA.

• **Hormones:** insulin (peptide, sugar level), thyroxine, adrenaline, testosterone, oestrogen (steroids).

Formula Sheet

Topic	Formula
Molarity / molality	M = n/V(L); m = n/W(kg)
Raoult's law	$p = x\,p°$; $(p° - p)/p° = x_2$
Boiling/freezing	$\Delta T_b = K_b m$; $\Delta T_f = K_f m$
Osmotic pressure	$\pi = CRT$
Cell potential	$E°_{cell} = E°_c - E°_a$
Nernst	$E = E° - (0.059/n) \log Q$
Gibbs energy	$\Delta G° = -nFE°$
Molar conductivity	$\Lambda_m = \kappa \times 1000/M$
First-order	$k = (2.303/t) \log([R]_0/[R])$; $t_{½} = 0.693/k$
Zero-order	$[R] = [R]_0 - kt$; $t_{½} = [R]_0/2k$
Arrhenius	k = A e^(−Ea/RT)
Magnetic moment	$\mu = \sqrt{[n(n+2)]}$ BM